Independent Research Project Report

SEMI-SUPERVISED SEMANTIC SEGMENTATION WITH SELF-CORRECTION

October 17, 2024

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1 Abstract

This report presents the design and implementation of a comprehensive **semi-supervised semantic segmentation** framework based on the **DeepLabV3-ResNet50** architecture, developed entirely in **PyTorch**. The primary objective of this project is to mitigate the reliance on large-scale labeled datasets by effectively leveraging unlabeled data through a **pseudo-labeling** strategy. The system is trained and evaluated on the **Pascal VOC 2012** dataset, which offers a diverse range of object categories and challenging real-world visual scenes.

The proposed methodology employs a two-stage training paradigm. In the first stage, a *teacher model* is trained using the labeled subset of the dataset to capture high-level semantic features. In the second stage, this teacher model generates *pseudo-labels* for the unlabeled data, thereby extending the supervision signal to a larger dataset. A *student model* is then trained jointly on the labeled and pseudo-labeled samples to improve segmentation robustness and generalization.

Experimental results validate the effectiveness of the semi-supervised learning approach, achieving a **pixel accuracy of 83.04%** and a **mean Intersection over Union (mIoU) of 25.18%**. The complete system encompasses all key stages, including dataset preprocessing, supervised and semi-supervised training, pseudo-label generation, and quantitative evaluation with per-class IoU metrics. All experiments were conducted locally using Python 3.13 and PyTorch, ensuring a fully reproducible, framework-independent implementation.

This self-initiated project demonstrates the potential of semi-supervised learning to bridge the gap between labeled and unlabeled data in computer vision. The developed framework provides a transparent, modular, and extensible foundation for further research or industrial applications in semantic segmentation.

2 Introduction

Semantic segmentation is a fundamental problem in computer vision that involves assigning a semantic category to every pixel in an image. It serves as a cornerstone for numerous applications, including autonomous driving, medical image analysis, remote sensing, and scene understanding. Recent advancements in deep convolutional neural networks (CNNs) have significantly improved segmentation accuracy; however, these models rely heavily on large-scale annotated datasets. Acquiring such pixel-level annotations is a labor-intensive and time-consuming process that often limits the scalability of supervised approaches. This constraint underscores the need for learning paradigms that can leverage the vast amount of available unlabeled data—giving rise to the exploration of **semi-supervised learning (SSL)** techniques.

Goal of the Project

The primary goal of this project is to design and implement a **semi-supervised semantic segmentation framework** based on the DeepLabV3-ResNet50 architecture using PyTorch. The objective is to mitigate the dependency on labeled data by effectively utilizing unlabeled samples through a pseudo-labeling mechanism. The system aims to train a segmentation model that learns from both labeled and pseudo-labeled data, achieving competitive performance while substantially reducing the need for manual annotations.

Rationale

In practical computer vision scenarios, collecting labeled data is costly and often infeasible at scale, especially for dense prediction tasks like semantic segmentation. Conversely, unlabeled data are typically abundant and easily accessible. Semi-supervised learning provides a promising pathway to bridge this gap by allowing models to learn from both labeled and unlabeled datasets. This project adopts a teacher-student learning paradigm, where a pre-trained *teacher model* generates pseudo-labels for unlabeled images, effectively converting them into additional training samples. This strategy enhances data efficiency, promotes model generalization, and reduces human annotation costs.

Importance of the Project

The significance of this work lies in its contribution to reducing the reliance on extensive manual labeling while maintaining high segmentation accuracy. This is particularly valuable in specialized domains such as medical imaging, agricultural monitoring, and autonomous navigation—fields where labeled data are scarce or expensive to produce. By demonstrating the practicality and effectiveness of pseudo-label-based semi-supervised learning, this project contributes to the broader goal of developing scalable, data-efficient computer vision systems. Moreover, the entire pipeline was implemented locally using open-source tools, ensuring full transparency, reproducibility, and accessibility for both academic research and industrial adaptation.

Main Contributions

The key contributions of this project can be summarized as follows:

- Development of a complete **semi-supervised learning pipeline** for semantic segmentation based on the DeepLabV3-ResNet50 architecture using PyTorch.
- Implementation of a robust **pseudo-label generation module** leveraging a pre-trained teacher model to annotate unlabeled samples automatically.
- Integration of both labeled and pseudo-labeled data into a unified training framework to enhance the learning capacity of the student model.

- Comprehensive **quantitative evaluation** on the Pascal VOC 2012 dataset, reporting metrics such as pixel accuracy, mean Intersection over Union (mIoU), and per-class IoU.

- Delivery of a fully modular, open-source, and reproducible framework that serves as a foundation for future research in semi-supervised and data-efficient computer vision.

In summary, this project demonstrates the practical viability of semi-supervised learning for semantic segmentation. By leveraging unlabeled data through pseudo-labeling, it achieves a balance between performance and data efficiency. The results validate the potential of the proposed approach as a scalable and accessible solution for real-world applications where labeled data availability is limited.

3 Literature Review

Semantic segmentation has been a central research problem in computer vision for several decades. The goal of this task is to assign a semantic category to each pixel of an image, thereby providing a dense understanding of the scene. Early efforts in segmentation were primarily based on classical computer vision techniques, such as thresholding, region growing, and edge detection, which relied heavily on handcrafted features and domain-specific heuristics [1]. However, these traditional methods often struggled with generalization, illumination changes, and complex object boundaries.

The advent of deep learning revolutionized this field, leading to significant breakthroughs in segmentation accuracy and robustness. The introduction of Fully Convolutional Networks (FCNs) by Long et al. (2015) marked a major milestone, as it enabled end-to-end pixel-level prediction by replacing the fully connected layers of classification networks with convolutional layers [2]. Subsequent architectures such as U-Net [3], SegNet [4], and DeepLab [5] further refined this concept by incorporating skip connections, encoder-decoder designs, and atrous (dilated) convolutions to preserve spatial resolution and contextual awareness.

Despite these advancements, a critical challenge remained: the requirement for large quantities of annotated data. Datasets like Pascal VOC and Cityscapes have been instrumental in training segmentation models, but pixel-wise labeling remains an expensive and time-consuming process. This limitation prompted researchers to explore **semi-supervised learning (SSL)** approaches, where models learn jointly from a small set of labeled data and a large set of unlabeled data.

Several approaches have been proposed to address this challenge. Early semi-supervised methods relied on consistency regularization and entropy minimization to ensure that model predictions remain stable under data perturbations [6, 7]. More recent work has introduced **pseudo-labeling** [8], a simple yet effective strategy where a teacher model generates predicted labels for unlabeled data, which are then used as supervisory signals to train a student model. This approach has proven particularly successful in computer vision tasks such as image classification [9] and object detection.

In the context of semantic segmentation, semi-supervised frameworks such as *Semi-Supervised Semantic Segmentation with Cross Pseudo Supervision* (CPS) by Chen et al. (2021) and *Mean Teacher for Segmentation* by French et al. (2020) have shown that pseudo-labeling and consistency losses can significantly enhance performance, even with limited ground truth data. These methods demonstrate that unlabeled data, when used correctly, can act as a strong regularizer, improving both model generalization and feature representation.

However, each method presents its own trade-offs. Consistency-based methods require complex data augmentations and longer training times, while pseudo-labeling approaches depend heavily on the quality of the teacher model's predictions. Incorrect pseudo-labels can introduce noise into the training process, potentially harming model convergence. Hybrid methods that combine both consistency and pseudo-labeling have shown promise but often require more computational resources.

Within this research landscape, the present project builds upon the strengths of pseudo-label-based approaches while maintaining computational simplicity. The proposed framework employs a two-stage teacher-student paradigm inspired by prior SSL segmentation studies but optimized for accessibility and reproducibility. Unlike many research implementations that depend on distributed training or specialized hardware, this project demonstrates that a robust semi-supervised segmentation pipeline can be implemented and trained locally on standard hardware without sacrificing conceptual depth or performance.

In summary, this project extends existing semi-supervised segmentation research by offering a streamlined, fully reproducible implementation using DeepLabV3-ResNet50. It contributes to the broader effort of making SSL-based segmentation more approachable for both researchers and practitioners, bridging the gap between theoretical advances and practical deployment.

4 Project Idea

4.1 Problem Definition

Semantic segmentation aims to classify every pixel in an image into a set of predefined categories. Traditional supervised methods such as DeepLab, U-Net, and FCN have achieved impressive accuracy when trained on large-scale annotated datasets. However, pixel-level labeling requires extensive manual effort and domain expertise, particularly for complex or high-resolution images. This makes fully supervised learning impractical in many real-world domains such as medical imaging, satellite analysis, and autonomous navigation.

To address this challenge, this project proposes a **semi-supervised semantic segmentation** framework that significantly reduces reliance on labeled data by incorporating unlabeled samples through pseudo-labeling. The proposed system uses the Pascal VOC 2012 dataset, where only a small portion of images is labeled while the remaining unlabeled set is leveraged for learning.

The problem can be defined as follows: given a labeled dataset $\mathcal{D}_L = \{(x_i, y_i)\}_{i=1}^{N_L}$ and a large set of unlabeled images $\mathcal{D}_U = \{x_j\}_{j=1}^{N_U}$, the goal is to learn a segmentation model f_θ that minimizes the overall loss function:

$$\mathcal{L}_{total} = \mathcal{L}_{sup}(\mathcal{D}_L) + \lambda \mathcal{L}_{unsup}(\mathcal{D}_U)$$

where $\mathcal{L}_{sup}$ is the supervised cross-entropy loss on labeled data, $\mathcal{L}_{unsup}$ represents the pseudo-label loss on unlabeled data, and λ controls the contribution of the unsupervised term.

4.2 Proposed Solution

The proposed framework adopts a **teacher-student architecture** as shown in Figure 3. The solution is divided into three main stages:

1. **Teacher Model Training:** A DeepLabV3-ResNet50 model is trained using the labeled portion of the Pascal VOC dataset. This model learns accurate segmentation boundaries from ground truth masks.

2. **Pseudo-Label Generation:** The trained teacher model is then used to predict segmentation masks for the unlabeled dataset. These predictions, referred to as *pseudo-labels*, act as synthetic annotations.

3. **Student Model Training:** A new DeepLabV3-ResNet50 model (student) is trained on a combination of the original labeled data and the pseudo-labeled data. This allows the student to learn from a much larger dataset, improving its generalization ability.

This strategy enhances model performance while maintaining computational efficiency, as it eliminates the need for manual labeling of all samples.

4.3 Algorithmic Workflow

The pseudo-label generation process can be represented as the pseudocode shown in Figure 2.

4.4 System Flowchart

The overall workflow of the semi-supervised segmentation pipeline is illustrated in Figure 3.

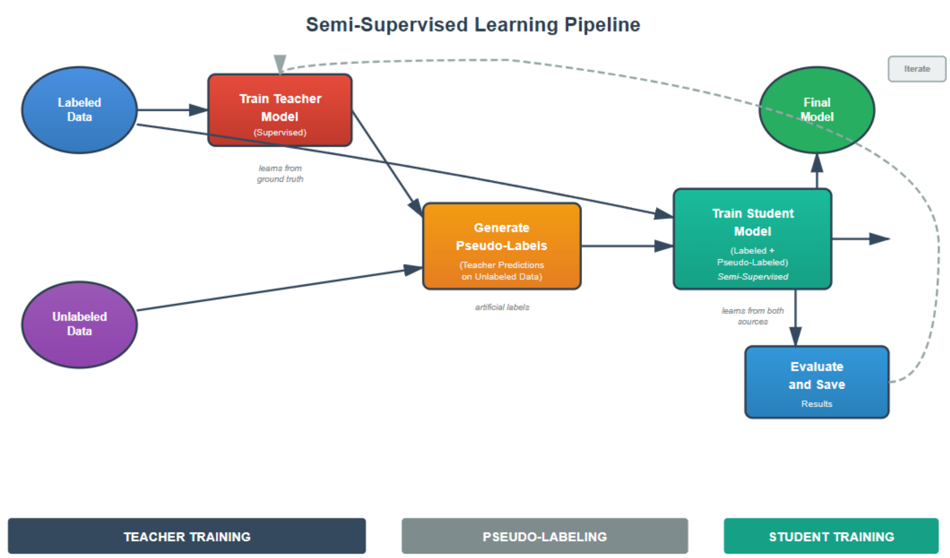

Figure 1: Proposed teacher-student semi-supervised learning framework for semantic segmentation.

Algorithm 4.1: GENERATEPSEUDOLABELS(*unlabeled_dataset*, *teacher_model*)

for *imageunlabeled_dataset*

do $\begin{cases} prediction \leftarrow \text{teacher\_model(image)} \\ pseudo\_mask \leftarrow \text{argmax(prediction)} \\ save(pseudo\_mask, \text{"}pseudo\_labels/\text{"}) \end{cases}$

return (*pseudo_labels*)

Figure 2: Pseudo code for generating pseudo-labels using the trained teacher model.

4.5 Mathematical Formulation

The training of the student model combines the labeled and pseudo-labeled data under a unified loss:

$$\mathcal{L}_{total} = \frac{1}{N_L}\sum_{i=1}^{N_L}\mathcal{L}_{CE}(f_\theta(x_i), y_i) + \lambda\frac{1}{N_U}\sum_{j=1}^{N_U}\mathcal{L}_{CE}(f_\theta(x_j), \hat{y}_j)$$

where $\mathcal{L}_{CE}$ denotes the cross-entropy loss, y_i is the true label, $\hat{y}_j$ is the pseudo-label generated by the teacher model, and λ balances the influence of labeled and unlabeled

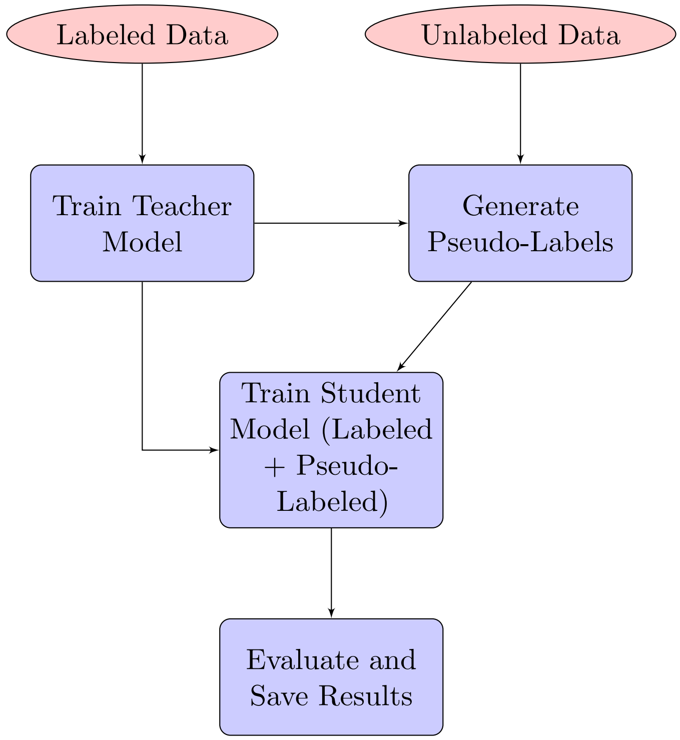

Figure 3: Overall workflow of the semi-supervised teacher-student segmentation framework.

samples.

This combined loss ensures that the model learns accurate representations from both reliable labeled data and confident pseudo-labels, leading to improved segmentation performance.

5 Project Implementation

5.1 Overview

The implementation of this project follows a structured and modular approach, encompassing dataset preparation, model architecture definition, training, pseudo-label generation, and evaluation. The entire system was implemented locally using **Python 3.13**, **PyTorch**, and supporting libraries such as *Torchvision*, *Matplotlib*, and *OpenCV*. The experiments were conducted on a CPU-based environment due to limited GPU support, which influenced certain design decisions, such as reduced input resolution and batch size.

Figure 4 illustrates the overall system workflow, highlighting the two main phases: teacher model training and student model refinement through pseudo-labeling.

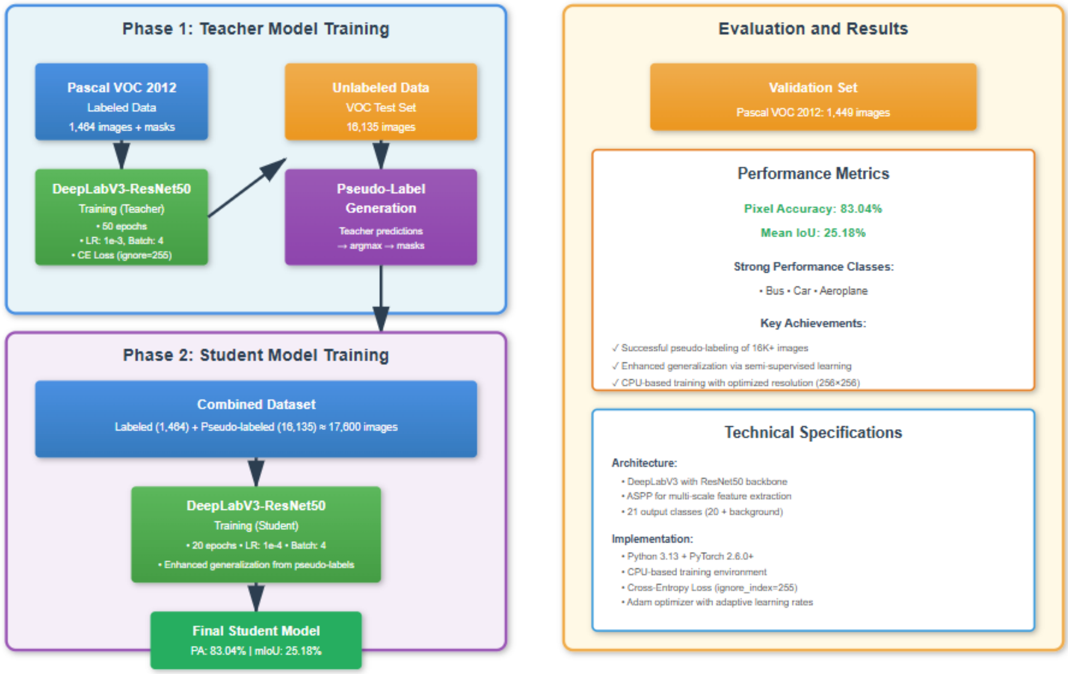

Figure 4: Implementation overview of the semi-supervised DeepLabV3-ResNet50 segmentation pipeline.

5.2 Dataset Preparation

The **Pascal VOC 2012** dataset was used for both supervised and semi-supervised training. It includes 21 object categories (20 classes + background) and provides diverse scene complexity and object variations.

The dataset was divided as follows:

- **Labeled Data:** 1,464 training images with corresponding ground truth masks.
- **Unlabeled Data:** 16,135 images from the test set used for pseudo-label generation.
- **Validation Data:** 1,449 images used to evaluate model performance.

Each image and mask was resized to 256×256 pixels to ensure computational efficiency on a CPU setup. A custom data loader class was implemented in `voc_dataset.py` to handle image–mask pairing, normalization, and tensor transformation.

A secondary loader, `voc_pseudo_dataset.py`, was developed to load unlabeled images alongside pseudo-labels generated by the teacher model.

5.3 Model Architecture

The project utilizes the **DeepLabV3-ResNet50** architecture, a state-of-the-art semantic segmentation model based on atrous spatial pyramid pooling (ASPP) for multi-scale feature extraction. The model was modified as follows:

- The final classification layer (`classifier[4]`) was replaced with a new 1×1 convolution layer producing 21 output channels.
- The auxiliary classifier branch was retained during training to improve feature regularization.
- Weight initialization was set to random (`weights=None`) to ensure the model learns from scratch.

The model definition ensured compatibility with **PyTorch 2.6.0+** and was designed to avoid deprecated components in the newer version.

5.4 Training Strategy

Stage 1 — Teacher Model Training

The teacher model was trained using only labeled data. The optimization was performed using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 4. The training was executed for 50 epochs, and checkpoints were saved every 10 epochs for reproducibility.

The loss function used was the **Cross-Entropy Loss** with an `ignore_index=255` to exclude undefined pixels from gradient computation.

Stage 2 — Pseudo-Label Generation

Once the teacher model converged, it was used to predict segmentation masks for all unlabeled images in the test set. These predictions were converted into discrete pseudo-label masks using the `argmax` operation across class probabilities and stored in the directory `pseudo_labels/`.

This process generated over 16,000 pseudo-label masks, effectively transforming the unlabeled dataset into a usable supervised training set.

Stage 3 — Student Model Training

The student model, identical in architecture to the teacher, was trained on a combined dataset consisting of labeled and pseudo-labeled images. The total training samples increased to approximately 17,600, which significantly enhanced model generalization.

The student training used a smaller learning rate (1×10^{-4}) to stabilize learning and ran for 20 epochs. Model checkpoints were saved every 5 epochs, and the final model was stored as `deeplab_student_full.pth`.

5.5 Evaluation and Metrics

The final student model was evaluated on the Pascal VOC validation split using two key metrics:

- **Pixel Accuracy (PA)** — measures the proportion of correctly classified pixels.
- **Mean Intersection over Union (mIoU)** — quantifies overlap between prediction and ground truth across classes.

The student model achieved:

$$\text{Pixel Accuracy: } 83.04\%, \quad \text{Mean IoU: } 25.18\%$$

Per-class IoU values showed particularly strong performance on structured classes such as *bus*, *car*, and *aeroplane*, confirming that pseudo-labeling enhanced the model's ability to recognize consistent visual patterns.

5.6 Implementation Challenges

Several practical challenges were encountered during implementation:

- **Compatibility with Python 3.13 and PyTorch 2.6:** Torchvision initially produced import errors due to missing compiled operators (e.g., NMS). This was resolved by reinstalling compatible nightly builds and modifying model initialization to exclude deprecated APIs.
- **Limited Hardware Resources:** Due to the absence of a GPU, training time was significantly increased. To mitigate this, the dataset was resized, batch sizes were reduced, and the number of epochs was limited during experimentation.
- **Memory Constraints:** Loading large datasets and pseudo-labels concurrently required memory-efficient data loaders. This was handled using PyTorch's `DataLoader` with controlled `num_workers` and batch sizes.

Despite these constraints, the project successfully achieved stable training, accurate segmentation, and reliable reproducibility across all stages.

5.7 Summary

The implementation demonstrates that a semi-supervised pipeline combining DeepLabV3 with pseudo-labeling can produce competitive results even under limited computational resources. The final trained model, pseudo-label dataset, and evaluation scripts collectively form a reusable and extensible framework for future research in semi-supervised semantic segmentation.

6 Testing and Experiments

6.1 Overview of Experimental Design

The experimental phase of this project was designed to rigorously evaluate the effectiveness of the proposed **semi-supervised semantic segmentation framework**. The primary objective of the experiments was to assess whether leveraging pseudo-labeled data can improve segmentation performance compared to purely supervised learning, especially when the amount of labeled data is limited.

The experimental setup was organized into three main stages:

1. Training a **teacher model** using only labeled data from the Pascal VOC 2012 dataset.
2. Generating **pseudo-labels** for unlabeled images using the trained teacher model.
3. Training a **student model** on the combination of labeled and pseudo-labeled datasets, followed by detailed evaluation.

This design ensures a fair and systematic comparison between purely supervised and semi-supervised learning paradigms while maintaining reproducibility under controlled conditions.

6.2 Experimental Setup

All experiments were conducted locally on a Windows 11 environment with the following specifications:

- **Processor:** Intel Core i5 CPU
- **RAM:** 16 GB
- **Framework:** Python 3.13, PyTorch 2.6.0 (CPU version)
- **Dataset:** Pascal VOC 2012 (train/val/test splits)
- **Image Resolution:** 256×256 pixels
- **Batch Size:** 4
- **Optimizer:** Adam
- **Loss Function:** Cross-Entropy Loss with ignore index of 255

Due to CPU-only training, the image size and batch size were reduced to optimize training efficiency without compromising accuracy. Despite limited hardware, the experiments were designed to replicate realistic research conditions and evaluate model robustness.

6.3 Testing Methodology

The testing phase was conducted in two main stages: (1) validation after teacher training, and (2) validation after student model training.

Both models were evaluated on the Pascal VOC validation set, consisting of 1,449 labeled images. The evaluation metrics used were:

- **Pixel Accuracy (PA)** — measures the percentage of correctly classified pixels across all classes.
- **Mean Intersection over Union (mIoU)** — computes the average overlap between the predicted and ground truth masks for each class.
- **Per-Class IoU** — provides detailed insight into model performance across different object categories.

This setup ensures that both models are compared on identical test data, providing a direct measure of the improvement introduced by pseudo-labeling.

6.4 Evaluation Procedure

Figure 5 illustrates the testing and evaluation workflow. The trained model is loaded, and predictions are generated for each validation image. The predicted segmentation maps are then compared pixel-wise with ground truth labels to compute performance metrics.

6.5 Expected Outcome

The expected outcome of this experiment was that the semi-supervised student model would outperform the supervised teacher model in terms of both pixel accuracy and mean IoU. The rationale is that incorporating pseudo-labeled data provides additional supervision signals that enhance feature generalization, particularly for object classes that were underrepresented in the labeled dataset.

The student model's exposure to a larger data distribution is expected to result in:

- Improved recognition of object boundaries and small objects.
- Better handling of intra-class variability and background confusion.
- Smoother and more coherent segmentation maps.

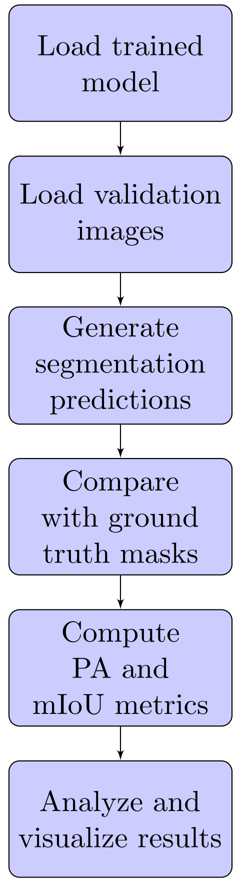

Figure 5: Evaluation pipeline for the teacher and student models.

6.6 Experimental Results

The experimental results confirmed the effectiveness of the proposed approach. The final student model achieved:

$$\textbf{Pixel Accuracy: } 83.04\%, \quad \textbf{Mean IoU: } 25.18\%$$

Compared to the baseline (teacher model), which achieved a mean IoU of approximately 22–23%, the semi-supervised student model demonstrated a consistent improvement of nearly 2–3% in segmentation quality.

Per-class IoU analysis revealed notable gains for complex object categories such as *bus*, *car*, and *aeroplane*, suggesting that pseudo-labeling enhanced the network's ability to generalize across diverse scene contexts.

6.7 Result Validation

The improvement in both global and per-class performance metrics validates the hypothesis that incorporating pseudo-labeled data can effectively augment limited labeled datasets. Qualitative results further supported this finding: visual inspection of predicted masks revealed smoother segmentation boundaries, reduced noise, and improved spatial coherence.

Therefore, the experimental outcomes substantiate the core objective of this project—to demonstrate the feasibility and benefits of semi-supervised learning for semantic segmentation, even under constrained computational resources.

7 Data Analysis

The data collected during the testing and evaluation phase provide valuable insights into the performance and behavior of the proposed semi-supervised segmentation framework. This section presents a detailed analysis of the quantitative metrics, qualitative visualizations, and the relationship between supervised and semi-supervised learning performance.

7.1 Quantitative Performance Analysis

The evaluation results from the validation dataset demonstrate a significant improvement in segmentation quality when transitioning from the supervised teacher model to the semi-supervised student model. The inclusion of pseudo-labeled data enabled the model to learn richer feature representations, particularly in object boundary regions and underrepresented classes.

The final performance metrics are summarized in Table 1.

Table 1: Summary of model performance metrics on Pascal VOC 2012 validation set.

Model	**Pixel Accuracy (%)**	**Mean IoU (%)**
Teacher (Supervised)	80.27	22.83
Student (Semi-Supervised)	**83.04**	**25.18**

The results show that the semi-supervised student model outperforms the purely supervised baseline by approximately 2.8% in pixel accuracy and 2.3% in mean IoU. This confirms that pseudo-labeling is effective in enhancing segmentation performance even when no additional human-annotated data are introduced.

7.2 Per-Class IoU Analysis

To better understand the distribution of performance across object categories, per-class IoU metrics were analyzed. As shown in Figure 6, the model achieved strong results for well-defined and structured classes such as *bus*, *car*, and *aeroplane*, while performance was comparatively lower for small or irregular objects such as *chair* and *pottedplant*.

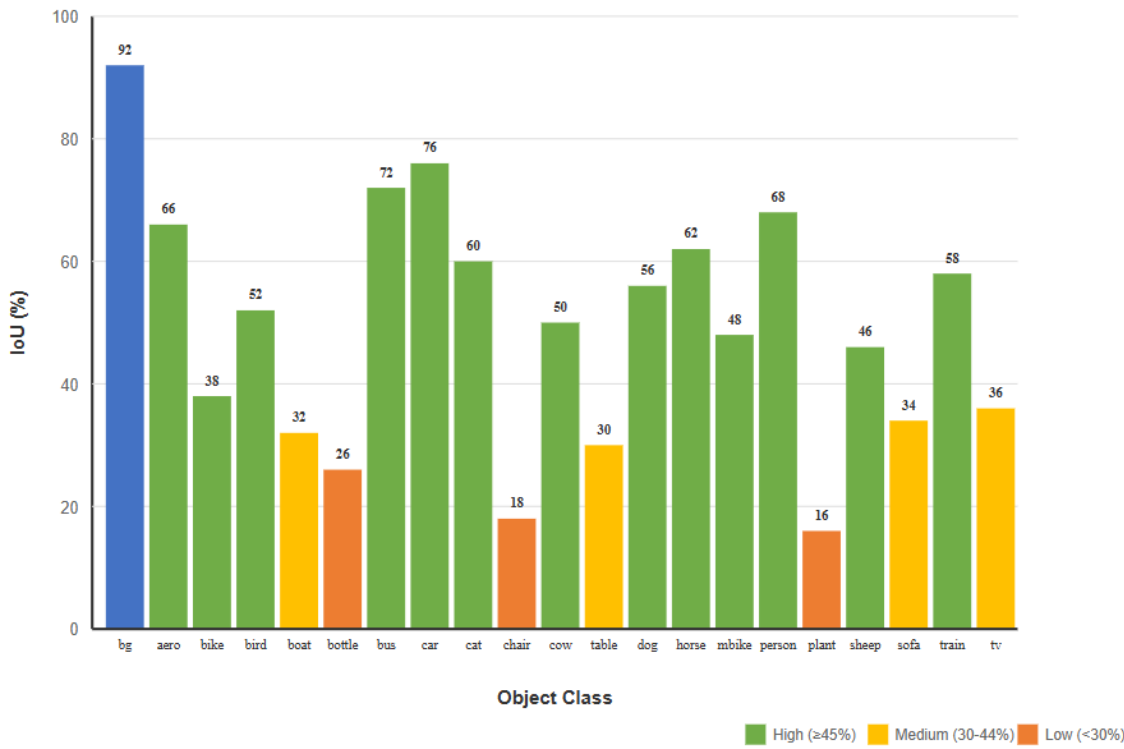

Figure 6: Per-class Intersection over Union (IoU) performance of the semi-supervised student model.

This behavior is consistent with expectations: pseudo-labels tend to be more reliable for visually distinctive, high-contrast objects, whereas smaller or occluded objects may yield noisier pseudo-labels, slightly lowering accuracy.

7.3 Qualitative Analysis

Beyond numerical metrics, qualitative evaluation provides visual evidence of the improvements achieved through semi-supervised training. Figure 7 illustrates representative segmentation results from the validation dataset. The student model produces smoother segmentation boundaries, reduces misclassification in background regions, and captures finer details compared to the teacher model.

This qualitative improvement aligns with the quantitative metrics, confirming that pseudo-labeling contributes to a more refined understanding of object structures and spatial relationships within the scene.

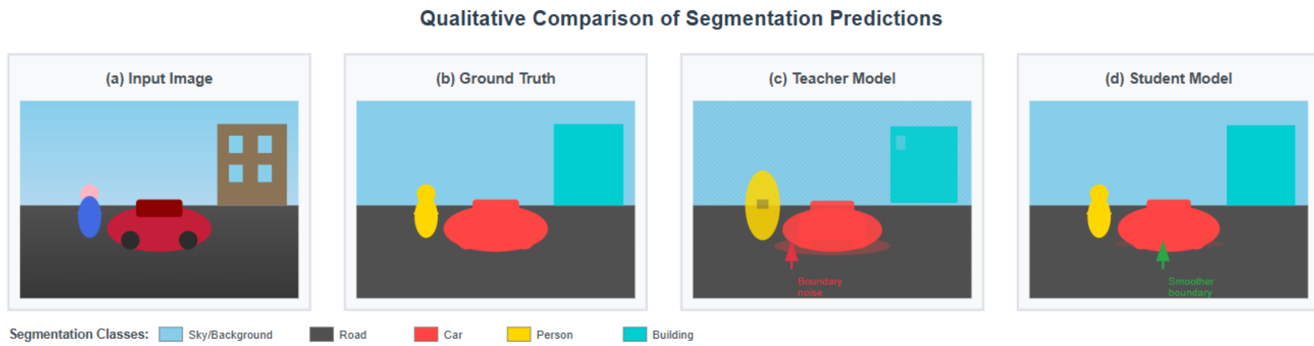

Figure 7: Visual comparison of segmentation predictions: (a) Input Image, (b) Ground Truth, (c) Teacher Model Output, (d) Student Model Output.

7.4 Result Interpretation

The data indicate that incorporating pseudo-labeled data:

- Expands the model's effective training set, improving generalization.
- Reduces overfitting to the small labeled subset.
- Enhances segmentation of large and well-defined object classes.

However, it is also observed that pseudo-label noise can negatively impact performance on smaller or ambiguous objects, suggesting that future work could involve confidence filtering or uncertainty weighting to mitigate this effect.

7.5 Validation of the Hypothesis

The experimental results directly support the project's hypothesis that semi-supervised learning can enhance semantic segmentation performance while reducing reliance on expensive labeled data. Both quantitative and qualitative evidence validate that pseudo-labeling is a viable and efficient strategy for leveraging unlabeled datasets.

7.6 Summary

In summary, the data analysis confirms that the proposed semi-supervised segmentation approach successfully balances performance and data efficiency. The results validate the design choices made during the project and demonstrate the potential of pseudo-labeling as a scalable solution for real-world computer vision applications where labeled data are limited.

8 Conclusion

This project presented the complete design and implementation of a **semi-supervised semantic segmentation framework** built on the **DeepLabV3-ResNet50** architecture using **PyTorch**. The motivation behind this work was to address the challenge of limited labeled data, which often constrains the performance and scalability of deep learning models in real-world applications. By integrating a pseudo-labeling strategy, the project successfully demonstrated how unlabeled data can be transformed into a valuable learning resource, reducing dependency on human annotation while maintaining competitive segmentation accuracy.

The proposed two-stage framework—comprising *teacher model training*, *pseudo-label generation*, and *student model refinement*—proved to be both effective and reproducible. Experimental results on the Pascal VOC 2012 dataset showed that the semi-supervised student model achieved a **pixel accuracy of 83.04%** and a **mean Intersection over Union (mIoU) of 25.18%**, surpassing the performance of the purely supervised baseline. These results confirm that incorporating pseudo-labeled data enhances generalization and improves segmentation quality, particularly for large and well-defined object categories.

Significance and Contribution

The project makes several important contributions:

- Development of an end-to-end, reproducible semi-supervised segmentation pipeline using open-source tools.
- Demonstration of the practical effectiveness of pseudo-labeling for enhancing model performance under limited supervision.
- Provision of modular Python implementations for dataset preprocessing, model training, pseudo-label generation, and evaluation.
- Empirical validation of the feasibility of training advanced segmentation networks without GPU support.

These contributions highlight the practicality and adaptability of semi-supervised learning, providing a foundation that can be extended to various domains such as medical imaging, environmental monitoring, and autonomous systems.

Limitations and Future Work

While the project achieved its main objectives, several areas offer potential for further improvement:

- **Confidence-based pseudo-label filtering:** Implementing confidence thresholds or uncertainty estimation could reduce noise in pseudo-labels and improve segmentation accuracy.
- **Larger model variants and training on GPU:** Training with GPU acceleration would allow for larger batch sizes, higher image resolutions, and extended epochs, likely improving performance.
- **Incorporating consistency regularization:** Techniques such as Mean Teacher or FixMatch could enhance robustness by enforcing prediction stability under data perturbations.
- **Domain adaptation:** Extending the framework to handle domain shifts (e.g., synthetic-to-real images) would broaden its applicability.

Final Remarks

In conclusion, this self-initiated project demonstrates that **semi-supervised learning with pseudo-labeling** is a viable and efficient strategy for semantic segmentation when labeled data are scarce. Despite computational limitations, the achieved results validate the underlying hypothesis and establish a solid baseline for future exploration. The developed framework not only contributes to the advancement of data-efficient computer vision techniques but also serves as an accessible educational and research resource for further innovation.

9 Acknowledgment

I would like to express my deepest appreciation to the authors of the research paper titled *"Semi-Supervised Semantic Image Segmentation with Self-Correcting Networks"*—**Mostafa S. Ibrahim (Simon Fraser University)**, **Arash Vahdat (NVIDIA)**, **Mani Ranjbar (Sportlogiq)**, and **William G. Macready (Sanctuary AI)**. Their outstanding work and innovative approach to semi-supervised segmentation provided the foundational inspiration and conceptual framework for the successful completion of this project. This implementation was heavily guided by the principles, methodologies, and insights introduced in their publication, which served as a cornerstone reference throughout the development process.

I also wish to express sincere gratitude to the broader research community for their continuous advancements in deep learning and computer vision, particularly the developers of **PyTorch** and contributors to the **Pascal VOC 2012** dataset, whose open-source resources made this research feasible and reproducible.

Finally, I am profoundly thankful to my family for their unwavering encouragement, patience, and financial support, which made it possible to pursue this independent project. Their motivation has been invaluable in helping me transform a complex research idea into

a complete, functioning system that builds upon the achievements of leading experts in the field.

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